

# POKHARA UNIVERSITY

Level: Bachelor  
 Programme: BE  
 Course: Data Communication (New)

Semester: Fall

Year : 2024  
 Full Marks : 100  
 Pass Marks : 45  
 Time : 3 hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

*Attempt all the questions.*

1. a) Explain the five essential components of a data communication system. How does each component contribute to successful communication? Explain with suitable diagram. 7
- b) What is serial and parallel data transmission? Differentiate between synchronous and asynchronous data transmission. 8
2. a) Check whether the given system is linear, time invariant and causal or not. 7
  - i.  $y(t) = \log x(t)$
  - ii.  $y(t) = t^2 x(t)$
- b) Differentiate between energy and power signal. Calculate whether ramp signal is power signal or energy. 8
3. a) Why Frame Relay is faster than X.25? Explain TCP/IP layers and compare it with OSI reference model. 8
- b) What is the purpose of cladding in an optical fibre? Explain the modes of propagation used in the optical fibre. 7
4. a) Define transmission impairments. And explain the causes of transmission impairments in details. 7
- b) If the 7 bit Hamming codeword received by a receiver is 1011011. Assuming the even parity, state whether the received codeword is correct or wrong. If wrong, decode the correct data word and locate the bit in error. 8

OR

Compress "Engineering" using Huffman codes and find its efficiency and redundancy.

5. a) What is ARQ? What are the two types of sliding window ARQ error control? Explain. 8

- b) What do you mean by multiplexing? How is TDM different from FDM? 7

6. a) Why do we need modulation need in data communication? Explain. 7

OR

Calculate the rate of a 500 baud 8-QAM signal with a constellation diagram.

- b) Represent the given bit stream 10011100001101 using 8
  - i. Polar NRZ
  - ii. Differential Manchester
  - iii. HDB3

7. Write short notes on: (Any two) 2×5

- a) Standards organizations
- b) Noise and its types
- c) HDLC protocol vs Point-to-Point protocol